

Software Requirement Specification (SRS)

ERP & CRM System for Share-Based Real Estate Development

1. Introduction

1.1 Purpose

The purpose of this SRS is to define the **functional, non-functional, and technical requirements** for a **custom ERP & CRM system** that will automate operations of share-based real estate development companies. The system will ensure **project management, financial transparency, governance, and shareholder communication**.

1.2 Scope

The system will provide:

- **CRM capabilities** for shareholder management, communication, and reporting
- **ERP capabilities** for project planning, finance, procurement, and construction tracking
- **Integration** between ERP and CRM modules for seamless operations
- **Web-based portals** for shareholders and management
- **Scalability** to handle multiple projects simultaneously

1.3 Definitions, Acronyms, Abbreviations

- **ERP:** Enterprise Resource Planning
- **CRM:** Customer Relationship Management
- **RBAC:** Role-Based Access Control
- **UAT:** User Acceptance Testing
- **SRS:** Software Requirement Specification

2. Overall Description

2.1 Product Perspective

The system will be **modular**, with four primary layers:

1. **CRM Layer:** Shareholder management, ledgers, notices, meetings
2. **ERP Layer:** Project, construction, finance, procurement
3. **Integration Layer:** Workflows, approval engine, role-based access
4. **Data & Security Layer:** Database, backup, compliance logging

2.2 Product Functions

- Shareholder onboarding and profile management
- Share allocation and phase-wise collection tracking
- Meeting and decision management

- Individual and consolidated ledgers
- Construction progress and project cost tracking
- Finance and accounting reports
- Procurement and vendor management
- Management fee calculation and reporting
- Dashboard and alerts for stakeholders
- Secure web/mobile access for shareholders

2.3 User Classes & Characteristics

User Class	Description	Permissions
Admin / Management	Company executives	Full access to ERP & CRM, create/approve transactions
Project Committee	Shareholder committee members	Approve project decisions, view reports
Accounts Staff	Finance and accounting personnel	Create entries, view cost, generate reports
Shareholders / Investors	Individual investors	View personal ledger, receive notices, track collections

2.4 Operating Environment

- Web-based and mobile-compatible interface
- Compatible with latest browsers (Chrome, Edge, Firefox)
- Backend hosted on centralized server/cloud
- Database: MySQL/PostgreSQL or SQL Server

2.5 Design & Implementation Constraints

- Compliance with Bangladesh IT rules & financial regulations
- Role-based access and secure authentication
- Audit trails for all actions and approvals
- Modular and scalable architecture

3. Functional Requirements

3.1 CRM Module

1. Shareholder registration and KYC capture
2. Share allocation and phase-wise payment tracking
3. Ledger generation (individual and project-wise)
4. Notices, communications, and meeting management

5. Shareholder portal (web & mobile)
6. Communication history logs
7. Reports and dashboards for analysis

3.2 ERP Module

1. Project & land management (acquisition, approvals, zoning)
2. Construction tracking (phases, milestones, costs)
3. Finance management (P&L, balance sheet, cash flow)
4. Management fee calculation (10–15%)
5. Procurement & vendor management
6. Cost allocation and reporting per project

3.3 Integration & Workflow

1. ERP ↔ CRM data synchronization
2. Approval workflows for project, finance, and procurement decisions
3. Role-based dashboards for each user type
4. Notification system for approvals, dues, and meetings

3.4 Data & Security

1. Centralized database with secure access
2. Daily backup and disaster recovery plan
3. Audit logs for compliance and traceability
4. Secure authentication (2FA optional)
5. Data encryption at rest and in transit

4. Non-Functional Requirements

Requirement	Description
Performance	The system should handle 10,000+ shareholder accounts and 50+ projects simultaneously
Availability	99.5% uptime for the web portal
Scalability	The system should support future expansion and multi-project management
Usability	Intuitive UI/UX for shareholders and staff
Security	Role-based access, encrypted storage, and audit trails
Compliance	Financial reporting aligned with Bangladesh accounting standards
Backup	Daily backups with a 30-day retention period

5. Interface Requirements

5.1 User Interfaces

- Dashboard for management with KPIs
- Shareholder portal with personal ledger and notices
- Form-based data entry for staff
- Interactive reports and charts

5.2 Hardware Interfaces

- Web server and database server
- Optional mobile devices for access

5.3 Software Interfaces

- MySQL/PostgreSQL/SQL Server database
- SMS/email gateways for notifications
- Optional integration with payment gateways

6. Data Requirements

- Shareholder personal & investment details
- Project data (land, cost, milestones)
- Construction phase & cost data
- Ledger entries and financial transactions
- Meeting and decision logs

7. System Architecture

- Modular ERP + CRM layers
- Centralized database
- API layer for integration
- Web/mobile front-end for accessibility
- RBAC for security and access control

(Can refer to the ASCII diagram from the previous architecture section)

8. Acceptance Criteria

- Successful completion of **User Acceptance Testing (UAT)**
- Shareholder portal is accessible and functional
- ERP & CRM modules fully integrated
- Reports and dashboards show **accurate and real-time data**
- Data migration verified
- Training completed and user manuals delivered

9. Implementation Phases

Phase	Activity	Duration
Phase 1	Requirement analysis & SRS finalization	1-2 weeks
Phase 2	System design & architecture	1 weeks
Phase 3	CRM development & testing	1-2 weeks
Phase 4	ERP module development	3-4 weeks
Phase 5	Integration, data migration, UAT	1-2 weeks
Phase 6	Training & Go-live support	2 weeks
Total Duration		2–3 months

10. Documentation & Training

- User manuals for ERP, CRM, and shareholder portal
- System administrator guide
- Training sessions for management, accounts, and staff

11. Maintenance & Support

- 12 months post-go-live support included
- AMC recommended at 18% annually for updates, bug fixes, and minor enhancements

12. Assumptions & Constraints

- Client will provide master data and timely approvals
- No major scope changes beyond SRS without contract amendment
- Hosting, SMS/email, and payment gateway costs excluded